Ruiyang Zhang

(+853) 63680758 | yc47931@um.edu.mo | ruiyang-061x.github.io
Ruiyang-061X | Ruiyang Zhang

Introduction

Ruiyang Zhang is a Ph.D. candidate at the University of Macau. He receives his B.S. in Computer Science and Technology from Fudan University.

His research focuses on enhancing the **reasoning** and **tool use capabilities** of **multimodal agents** through reinforcement learning. He previously worked on exploring the uncertainty of large multimodal models and on 3D object detection in unsupervised settings.

Education

University of Macau *Aug. 2024 -*
Ph.D. in Faculty of Science and Technology Macau SAR
Advisor: **Prof. Zhedong Zheng**

Fudan University *Sept. 2017 - Jun. 2021*
B.S. in Computer Science and Technology Shanghai, China

Research Experience

Research Intern @ Ant Group, Ling Team *June 2026 -*
Mentor: Qingpei Guo Beijing, China

- Topics: Exploring and developing general pipeline for constructing ultra-long-horizon ability
- Project: Marathoner
- Responsibilities: Propose a comprehensive post-training pipeline for constructing ultra-long-horizon execution ability into base model, including Ultra-Long-Horizon Task Synthesis, Rejection Sampling Finetuning, and Reinforcement Learning. For data synthesis, we also propose Multi-Task Chaining to produce even more challenging tasks. For RL, we also introduce Later Stage Bonus Reward, which explicitly rewards valuable operation during later stage of execution, facilitating more effective ultra-long-horizon ability.

Research Intern @ IDEA *Octo. 2025 - Mar. 2026*
Mentor: Dr. Yiyan Qi Shenzhen, China

- Topics: Developing long-horizon multimodal search agent via reinforcement learning (RL)
- Project: VSearcher (**NeurIPS'26**)
- Responsibilities: (1) Build the RL infrastructure required to develop multi-turn tool-using multimodal search agent, including tool implementation, caching mechanism, and RL framework adaptation. (2) Build long-horizon, multi-turn multimodal search agent in real-world web environment via reinforcement learning. Propose a systematic post-training framework for multimodal search agents, including Iterative Injection-based Data Synthesis, Rejection Sampling Fine-tuning, and RL.

Research Intern @ Shanghai AI Lab *May 2025 - Sept. 2025*
Mentor: Dr. Dongzhan Zhou Shanghai, China

- Topics: Leveraging RL to incentivize sketch-style reasoning in large multimodal models
- Project: SketchThinker-R1 (**ICLR'26**)
- Responsibilities: (i) Devise Sketch-Mode Cold Start to instill initial sketch-style reasoning into base

multimodal model; train SketchJudge reward model to pose supervision on thinking style during the RL stage; conduct Sketch-Thinking Reinforcement Learning to further generalize sketch-style reasoning. (ii) Evaluation on benchmarks across various domains, showing that SketchThinker-R1 achieves over 64% reduction in thinking cost without compromising answer accuracy.

Selected Award

- Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) 2019
- Third Class Scholarship for Outstanding Students, Fudan University 2018 & 2019
- First Prize, Chinese Mathematical Olympiad in Jiangsu Province (CMO) 2016
- First Prize, National Olympiad in Informatics in Jiangsu Province (NOIP) 2014 & 2015

Publication

1. **[Preprint] Marathoner: Ultra-Long-Horizon Autonomous Intelligence**
   Ruiyang Zhang, Jinpeng Ou, Yifan Xie, Jingang Zhou, Lirui Pan, Qingpei Guo, Zhedong Zheng
   [Paper]
   - Providing a comprehensive receipt for building ultra-long-horizon ability into vanilla base model.

2. **[NeurIPS 2026] VSearcher: Long-Horizon Multimodal Search Agent via Reinforcement Learning**
   Ruiyang Zhang, Qianguo Sun, Chao Song, Yiyan Qi, Zhedong Zheng
   [Paper] | **[Code]**
   - Training multi-turn tool use multimodal search agent in real-world web environment via reinforcement learning.

3. **[ICLR 2026] SketchThinker-R1: Towards Efficient Sketch-Style Reasoning in Large Multimodal Models**
   Ruiyang Zhang*, Dongzhan Zhou*, Zhedong Zheng
   [Paper] | **[Code]**
   - Incentivize sketch-style reasoning in large multimodal models to improve reasoning efficiency.

4. **[ICCV 2025] Harnessing Uncertainty-aware Bounding Boxes for Unsupervised 3D Object Detection**
   Ruiyang Zhang, Hu Zhang, Hang Yu, Zhedong Zheng
   [Paper] | **[Code]**
   - Aim to mitigate the negative impact of noisy pseudo boxes.

5. **[ECCV 2024] Approaching Outside: Scaling Unsupervised 3D Object Detection from 2D Scene [40+ Github Stars]**
   Ruiyang Zhang, Hu Zhang, Hang Yu, Zhedong Zheng
   [Paper] | **[Code]**
   - Enhance detection of far and small objects via LiDAR and 2D data fusion.

6. **[EMNLP 2026] Uncertainty-o: One Model-agnostic Framework for Unveiling Epistemic Uncertainty in Large Multimodal Models**
   Ruiyang Zhang, Hu Zhang, Hao Fei, Zhedong Zheng
   [Paper] | **[Code]** | **[Website]**
   - Applicable to both large comprehension model and large generation model.

7. **[Preprint] VL-Uncertainty: Detecting Hallucination in Large Vision-Language Model via Uncertainty Estimation [80+ Citations]**
   Ruiyang Zhang, Hu Zhang, Zhedong Zheng
   [Paper] | **[Code]** | **[Website]**
   - Preliminary attempt for understanding uncertainty in LVLM.

Work Experience

Meituan *Jul. 2021 - Jun. 2023*
Backend Development Engineer Shanghai, China

Open Source

1. [**Awesome List**] **Awesome-MLLM-Uncertainty** [**50+ Github Stars**] [**Code**]
   - Curated list of papers on the uncertainty in multi-modal large language model (MLLM).
2. [**Awesome List**] **Awesome-MLLM-Reasoning** [**Code**]
   - Curated list about reasoning abilitiy of MLLM, including OpenAI o1, OpenAI o3-mini, and Slow-Thinking.